Generative AI as a Study Coach in First-Year ICT Education

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***Abstract*—This study addresses the question of how far a generative AI study coach within first-year ICT education at the Amsterdam University of Applied Sciences can contribute to transparent and accountable learning, without critical thinking and independent performance dropping out of sight. In a short time, generative AI has become part of students' daily study behaviour, while the programme still has to be able to establish what students understand, substantiate and apply on their own. The study combines a literature review with practice-based research: a survey among students, baseline and final measurements within Business IT & Management, Cyber Security and Fusion, feedback sessions in class, conversations with people involved and an analysis of historical grades over 2019 to 2025. The main conclusion is that a generative AI study coach at this point is associated above all with a more visible learning process, more reflection and stronger checking of AI output. The approach is therefore useful mainly as a didactic means for transparency and self-regulation. For proven higher learning performance, the evidence is not yet sufficient. The advice is to keep measuring with a sharper definition of the task for each teaching context.**

***Index Terms*—generative AI, study coach, cognitive offloading, AI literacy, self-regulation, higher education**

Note on this version. This paper is based on the publication version of the graduation report; the original version was submitted to and assessed by the Amsterdam University of Applied Sciences. Results are shown at group level only, in Figures 5.1, 5.2 and 7.4 to 7.6, with the same figures that were presented at ICERI2026. No data of individual students is included. Other figures from the original report are left out, as noted in the text.

1. Introduction

1.1 Organisational context

This research takes place within the Amsterdam University of Applied Sciences, in the ICT bachelor's programme of the Faculty of Digital Media and Creative Industry. The HvA is a large university of applied sciences with a broad range of technical programmes and has, in recent years, invested in its own safe AI environment for students and staff through HvA AI Chat.

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Since the 2025-2026 academic year, the ICT programme has worked with a combined first year in which first-year students from different specialisations follow education together. Within this first year, students can get to know the specialisations that also return in this research: Business IT & Management, Cyber Security and Fusion. The education is built around sprint and project teaching, with regular check-ins, feedback sessions and peer review.

Within that context, this research works together with the first-year teachers and with the developers of HvA AI Chat. The use of a constrained study coach touches directly on the way the first year wants to make its learning outcomes visible. The definition of the problem, set out in the following sections, also follows from that organisation.

Several people are involved in this research, each with their own role.

Research and execution: Hugo van der Geest, graduating ICT student.

Supervision: Martin Spoor supervises the graduation project on behalf of the HvA, and Stijn Knigge is the second supervisor. Philip Lafeber supervises from practice and also teaches as a BIM lecturer.

BIM lecturers: Maarten Post, Rob Strober, Sybe Dijkstra and Jelle Kaptijn.

Cyber Security lecturers: Maaike Valk, Enrico de Vries and Maarten Bonsema.

Fusion lecturers: Bas Pijls, Jorrit de Haas, Pieta Harsveld and Remco van Swieten.

Conversation partners and experts: Ron Mathot on fraud and assessment, Constance Bleeker on educational innovation, Pascal Wiggers and Pieta Harsveld on ethics, Jan Berends and Jan-Hein Eggers on research methodology, Izaak Dekker on literature and the design of experiments, and Coline Pannier, Michal Kozdon and Lissana Flens on HvA AI Chat and the frameworks from ICTS.

1.2 Background

On 10 March 2026, the research diary recorded that the ICT first year was working on the design of education for the 2026-2027 academic year. That design centred on learning outcomes, learning activities and assessment. This meant that,

within the programme, there was already a concrete teaching context in which the question became relevant of how the use of generative AI relates to what students have to know and be able to do independently (see Appendix B).

That tension connects to the literature on cognitive offloading. Risko and Gilbert use that term for outsourcing thinking to external aids. For this research, that is relevant because generative AI can not only give answers, but can also take over intermediate steps, wordings and choices that are normally part of the learning process (Risko & Gilbert, 2016).

In the research diary, that didactic question was then also named in the practice context. Maarten Bonsema, a Cyber Security teacher, linked the use of AI on 25 March 2026 directly to the question of which learning objectives are central in an assignment; only then can it be decided which work can and cannot be left to AI (Bonsema, 2026) (see Appendix H3). A day later, a conversation with Lafeber recorded that the term coach was chosen deliberately, because it is not the chatbot but the teacher who remains responsible for learning and for achieving the learning objectives (see Appendix H4). Against this background, this research focuses on the use of a generative AI study coach within the ICT first year.

1.3 Problem definition

1.3.1 Teaching context and assessability: Within the ICT first year at the Amsterdam University of Applied Sciences, it has to remain visible what students can analyse, substantiate and apply independently. For the HvA, that is not an abstract principle but a condition for good teaching, reliable assessment and a defensible link between learning outcomes, learning activities and assessment. In conversations about educational innovation, assessment and the design of the first year, it emerged that this coherence is a clear concern within the HvA, precisely because the programme has to be able to account for what students master themselves and how that is established. When that view diminishes, it affects not only individual students but also the quality and credibility of the teaching process (see Appendix H2) (NPULS, 2025; Bleeker, 2026).

That teaching context was also sharpened internally in January 2026 in the HBO-ICT position paper AI in the curriculum. It sets out a two-track approach in which generative AI is given room for learning and feedback, but is deliberately bounded in qualifying contexts so that statements about individual competence remain valid and lasting knowledge is not outsourced too early (Berends, Flinkenflögel, & Van Swieten, 2026).

A second internal HvA line makes that context sharper still. The discussion note on the compulsory use of HvA AI Chat sets out that, in some course units, generative AI is not only supportive but can be part of the competence a programme deliberately wants to develop. At the same time, such a requirement stays tied to learning objectives, final qualifications and the programme profile, not to the tool itself (Hogeschool van Amsterdam, z.d.).

1.3.2 Generative AI as an existing reality in education: Generative AI is by now part of students' study behaviour. In the first survey, some of the students surveyed reported using AI weekly or daily. In surveys, baseline measurements and feedback conversations, it also emerged that AI is used for explanation, feedback, improving text, sparring and speeding up assignments. For the programme, the use of AI is therefore no longer an exception but an existing practice that has to be taken into account in teaching and assessment (see Appendix F1 and Appendix B).

1.3.3 The risk of cognitive offloading: Risko and Gilbert describe cognitive offloading as the outsourcing of mental actions to external aids. In a teaching context, this is not only about memory support, but also about shifting analysing, structuring, comparing and formulating to a tool. That is exactly what becomes problematic when, as a result, students have to explain less of the thinking steps they have taken. In the first survey, the exploratory feedback sessions and the literature review, it became visible that generative AI can work both ways: targeted use can help with explanation, planning and feedback, while untargeted use can lead to thinking steps, considerations and justifications dropping out of sight.

1.3.4 Why this is a concrete problem for the HvA: For the HvA, cognitive offloading becomes a concrete teaching problem as soon as generative AI not only supports, but also takes over the analysis, formulation and justification that the first year is meant to practise. As a result, it becomes less visible which thinking steps the student has taken themselves. This affects the quality of the learning process, the reliability of supervision and assessment, and the programme's responsibility to assess learning outcomes in a defensible way. The heart of the issue therefore does not lie in a general ban on AI, but in the question of how the use of AI can be didactically bounded, made visible and kept checkable.

1.3.5 Scope of the problem: The scope of the problem is set out in this research using the 5W approach:

1. Who: first-year students within the ICT first year, the teachers involved and the programme as the assessing context.
2. What: through broad but not always guided use of generative AI, it becomes less visible which thinking steps, choices and products still come about independently from students.
3. Where: within the ICT first year, in lessons, assignments, feedback sessions and forms of assessment.
4. When: in current teaching practice, where generative AI is already part of students' daily study behaviour.
5. Why: the programme has to be able to establish whether students actually develop, master and account for the intended learning outcomes themselves.

This scope deliberately focuses on the ICT first year. It is precisely in this phase that broad project contexts, first-year skills, orientation towards several IT domains and the assessability of basic competences come together. As a result, the risk of the invisible outsourcing of thinking can be studied most sharply, both methodologically and didactically, in the first year.

1.3.6 Causes and the core of the problem: This problem analysis is worked out as a root cause analysis in the form of an Ishikawa or fishbone approach. That method helps to avoid getting stuck at symptoms and instead to order possible causes in relation to one another. In this thesis, the causes of the problem are placed in five main groups:(ICT Research Methods, z.d.)

1. Student behaviour: students often use generative AI in a practical way for speed, convenience and direct help, so that the line between support and the takeover of thinking easily blurs.

2. Didactics: shared frameworks for learning-oriented use are still being developed, so that for each assignment and each lesson it is not always clear what role AI may and may not play.

3. Process and accountability: the use of AI is not yet recorded visibly in every assignment or form of assessment, while teachers do need that information to follow learning processes.

4. Knowledge and skills: AI literacy, including critical assessment, source checking and ethical considerations, is not yet present at the same level among all students.

5. Organisational goals: the programme has to prepare students for professional use of AI and, at the same time, make sure that independent critical thinking and assessable performance do not fade into the background.

The core of the problem for the HvA is therefore not that students use generative AI, but that this use, without clear didactic and organisational limits, puts pressure on the relationship between learning outcomes, learning activities and assessment. When it is insufficiently visible what students understand, formulate and account for themselves, it becomes harder to assess reliably, to supervise in a targeted way and to safeguard the quality of the teaching. From that broader HvA issue, this research therefore focuses on a bounded sub-problem within the ICT first year: under what conditions a bounded form of AI use can support students, without critical thinking, transparency and independent performance dropping further out of sight (see Appendix H1 and H2) (NPULS, 2025; Mathot, 2026; Bleeker, 2026).

1.4 Objective

The aim of this practice-based research is to make clear, for the Amsterdam University of Applied Sciences, under what didactic conditions generative AI can be used responsibly within first-year ICT education. The scope is deliberately limited to the first year, because it is precisely there that the combination of first-year skills, broad project assignments, orientation towards several domains and early assessment moments most clearly shows which thinking steps students still carry out independently.

In this research, a generative AI study coach means an AI environment that does not directly give complete answers or solutions, but prompts students towards their own reasoning through counter-questions, hints and step-by-step guidance. The objective thereby connects to the literature on cognitive offloading, which centres on outsourcing mental processes to external aids, and to the NPULS framework, in which critical, conscious and responsible use of AI is part of AI literacy (NPULS, 2025) (Risko & Gilbert, 2016).

The relevance for the HvA lies in the tension between two educational goals: preparing students for professional use of AI and, at the same time, safeguarding assessment, independence and durable knowledge building (see Appendix H2). For the client, this means that the research has to provide a well-founded basis for the further didactic use of a generative AI study coach within first-year ICT education. The intended outcome is therefore a guiding recommendation for design choices and usage guidelines that match the educational goals and preconditions of the HvA (NPULS, 2025; Risko & Gilbert, 2016; Bleeker, 2026).

1.5 Main question and sub-questions

This research is built around the following main question:

> To what extent does an HvA generative AI study coach help to account transparently for learning performance, without the critical thinking of first-year ICT students beginning to suffer as a result?

To answer the main question in a structured way, five sub-questions are used. In Chapter 3, each is linked to a specific method, and in Chapters 4 to 7 they are answered.

> Sub-question 1: What is meant in this research by critical thinking in relation to cognitive offloading and mental friction?
> Sub-question 2: How do first-year ICT students currently experience and use generative AI in the daily practice of their learning process?
> Sub-question 3: Under what didactic conditions can a constrained AI coach function as study support?
> Sub-question 4: Is a trend visible in historical learning performance since the broad rise of generative AI, and how cautiously should that trend be interpreted?
> Sub-question 5: What measurable difference in learning performance, independence and visibility of the learning process is visible in practice between an open and a constrained condition?

1.6 Definition of key concepts

Four concepts are central to this research: critical thinking, cognitive offloading, mental friction and the generative AI study coach. These concepts guide the analysis of the learning process, the interpretation of AI use and the assessment of independence. The working definitions are given beforehand in the glossary; in Chapter 2 these concepts are worked out and substantiated theoretically. Within this thesis, critical thinking is therefore not understood as avoiding AI, but as keeping one's own choices, reasoning and intermediate steps visible and checkable during the learning process.

1.7 Reading guide

This graduation report is built up from a number of fixed steps. After this introduction comes the theoretical framework. That second chapter deals with the theory behind cognitive offloading and the learning mechanisms. Chapter three describes the methodology, including the design of the A/B test. Chapters four to seven answer the sub-questions in a fixed structure: introduction, analysis and sub-conclusion. Chapter eight contains the final conclusion. Chapter nine contains the recommendations for the programme. Chapter ten is the list of references.

2. Theoretical framework

2.1 Cognitive offloading and mental friction

2.1.1 Critical thinking as a starting point: Critical thinking is the central concept in this research. It is not only about reproducing knowledge, but about independently analysing, assessing and substantiating information in more complex situations. Anghel et al. describe this as the ability to reason in realistic dilemmas. For this research, that is relevant, because in programming, analysis and design tasks first-year ICT students not only have to give an outcome, but also have to be able to account for their choices.

2.1.2 Why generative AI makes this topic more urgent: The rapid rise of generative AI makes this topic more urgent. ChatGPT and similar systems give immediate explanations, examples, summaries and alternative wordings, so that students can move through a task more quickly. That is not necessarily unfavourable. Suriano et al. found, among 213 Italian students, a connection between engagement with ChatGPT, knowledge about AI and performance on a critical reasoning assessment. Guo and Lee also describe, in chemistry education, that targeted use of ChatGPT can help students formulate better questions and analyse information more deeply (Miao & Holmes, 2023; OpenAI, 2022).(2023)

At the same time, not all studies point in the same direction. Kosar et al. found, in a programming course for novice programmers, no significant difference in academic performance between students with and without ChatGPT. The literature therefore does not show a clear picture in which generative AI automatically works out positively or negatively for learning (Guo & Lee, 2023; Kosar, Ostojić, Liu, & Mernik, 2024; Suriano, Plebe, Acciai, & Fabio, 2025).

2.1.3 Cognitive offloading as a risk: For that very reason, it is not only the presence of AI that matters, but above all the way students work with it. Risko and Gilbert describe cognitive offloading as the outsourcing of mental actions to external aids. With generative AI, this goes further than just looking up or storing information. The tool can also summarise, structure, rewrite and generate provisional reasoning. As a result, there is a risk that students arrive at an answer more quickly, but analyse, compare and put into words less actively.

That risk is also visible in more recent research. Barcaui showed, in a study with 120 students, that the group that used ChatGPT as a study aid scored lower on knowledge retention after 45 days than the group that studied in the traditional way: 57.5 per cent against 68.5 per cent.

A systematic review by Li, Cui and Hagedorn, based on 67 studies from 2022 to 2025, also shows that in higher education ChatGPT is more often positively associated with creative outcomes than with critical thinking. For this research, that is relevant, because faster idea generation or richer output does not yet mean that critical thinking also increases.

2.1.4 Mental friction, guidance and self-regulation: The literature also addresses didactic guidance and self-regulation. Lee et al. show, in blended learning, that a guidance mechanism, with hints and steering questions instead of direct solutions, works out more favourably for self-regulation, knowledge construction and higher-order thinking. Miao and Holmes also stress that generative AI in education should not be introduced neutrally, but should be pedagogically designed and used in a human-centred way.

For higher education, this means that the question is not only whether students use AI, but also whether the educational design still forces them to make intermediate steps explicit, check output and form their own judgement. That is exactly the point at which critical thinking and generative AI meet.

2.1.5 Relation to AI literacy: AI literacy is directly connected to this. The NPULS framework describes AI literacy as a combination of knowledge, skills and attitude, with attention to critical and ethical conduct. Ng et al. work this out further in four dimensions: understanding AI, applying AI, evaluating and creating AI, and recognising ethical issues. That evaluation in particular is relevant here. Students should not only know how to use a chatbot, but also be able to assess the origin, reliability, limitations and possible bias of the output. Without that step, the use of AI quickly shifts from support to a replacement for one's own thinking (Miao & Holmes, 2023; Ng, Leung, Chu, & Qiao, 2021; Risko & Gilbert, 2016).

2.1.6 Meaning for this research: From this literature, critical thinking is understood in this research as dealing independently and with reasoned argument with information, choices and problem steps in an AI-rich learning environment. Cognitive offloading and mental friction function as explanatory concepts, not as the central end goal of the research. Constrained use of generative AI is defined in this thesis as a didactically bounded form of AI use in which the tool does not primarily produce answers, but prompts the student, through hints, counter-questions and structure, to analyse, weigh up and formulate themselves. It is precisely in that tension between support and outsourcing that the heart of this research into first-year ICT students at the HvA lies.

2.2 AI literacy, accountability and assessment

2.2.1 AI literacy as an educational framework: Within higher education, the use of generative AI is not only a technical issue, but also a question of which knowledge, skills and attitudes students need to use such systems responsibly. Ng et al. describe AI literacy as more than operational skill: students should be able to understand, apply, evaluate and place AI within ethical issues. The Dutch AI GO! framework

connects to this and defines AI literacy as an interplay of knowledge, skills, attitudes and ethical awareness. That framework is moreover not only descriptive but evidence-informed: it is built up from a review of existing academic literature and supplemented with practice feedback from education professionals in senior secondary vocational, higher professional and university education (NPULS, 2025). AI literacy is therefore not a loose term in this thesis, but a well-founded framework for interpreting the educational use of AI.

An additional framework that fits well here is the AI Fluency Framework by Dakan and Feller. They describe AI fluency as the ability to work with AI effectively, efficiently, ethically and safely across different forms of human-AI interaction. They distinguish four core skills: delegation, description, discernment and diligence. For this thesis, that is relevant because the framework is not only about prompting technique, but precisely about the division of tasks, quality control of output and the responsible embedding of AI use in a learning context.

2.2.2 What AI literacy asks of students: For first-year ICT students, AI literacy means that they not only know how a chatbot can be used technically, but also when that use is appropriate, what limitations a model has and how the quality of the output should be assessed. With generative AI in particular, that is relevant, because these systems can formulate convincingly without the content necessarily being correct, complete or checkable. Miao and Holmes therefore stress that generative AI in education can only be used responsibly when institutions pay attention to privacy and ethical considerations. Along the same lines, Ng et al. argue that evaluating and acting ethically are not a side issue, but core parts of AI literacy. For this research, that matters, because the risk lies not only in faulty output, but also in the gradual shift of thinking from the student to the system.

2.2.3 Accountability and self-regulation: That shift touches directly on self-regulation and accountability. When students use AI as support, they have to keep making choices about what they outsource, what they check themselves and how they process AI output in their own learning. Lee et al. show that guiding AI support with hints and targeted questions works out more favourably for self-regulation, knowledge construction and higher-order thinking than a form of use in which answers become directly available. AI literacy therefore also calls for a reflective attitude. Students should not only be able to work with AI, but also be able to explain why they choose a particular use and which contribution of their own remains visible in it. In this thesis, AI literacy is therefore understood as critical, purposeful and transparent use of generative AI, in which the student keeps responsibility for the substantive choices and for accounting for them (Ng, Leung, Chu, & Qiao, 2021; NPULS, 2025).

Mutsaarts translates this line for Dutch pupils and students into the KIES framework: Choose, Instruct, Evaluate and Ground rules. Two elements in particular connect to the issue in this thesis. First, the use of AI is not treated as a choice for a whole assignment, but for each subtask. Second, the human-AI-human principle is central, with AI output seen as raw material and not as an end product. KIES thereby connects to the emphasis on accountability, because students should not only be able to work with AI, but also be able to explain which thinking steps they have taken themselves.

2.2.4 Assessment and examination: From that perspective, AI literacy cannot be seen apart from assessment and examination. The NPULS vision on assessment, examination and AI argues that institutions have to work towards AI-aware assessment practices that are valid, fair and aimed at learning objectives, and that at the same time strengthen assessment integrity, self-regulation and AI literacy (NPULS, 2025). The central question is then not only whether the use of AI amounts to fraud, but above all which learning process and which learning outcomes are being assessed. When a test only values the end product, it becomes less visible how much analysis, selection and reasoning the student carried out themselves. With the use of AI, that question becomes even more urgent, because a convincing answer does not automatically mean that there is independent understanding. Within this research, the emphasis therefore does not lie on designing new forms of assessment, but on describing the conditions under which existing assessment and examination can remain AI-aware and checkable.

Recent HvA guidelines make this distinction more concrete. The working document on marking and giving feedback with AI makes clear that AI can at most be used in a supporting role for formative feedback on practice work, while the summative assessment of student work at the HvA is not carried out with AI. The same document also rejects AI detection tools as a basis for assessment, because they do not give reliable statements about student use (Hogeschool van Amsterdam, z.d.).

2.2.5 Link with the practice exploration: That link between use and assessment also emerged in the exploratory phase of this practice-based research. In conversations about plagiarism, assessability and the research design, attention shifted again and again from the question of whether AI is used at all to the question of which learning process, which degree of independence and which accountability have to remain visible when students use generative AI. In this way, the practice exploration connected to the literature: it is not the simple distinction between using AI or not that is central, but the question of under what conditions the use of AI still serves learning (see Appendix H1) (NPULS, 2025; Mathot, 2026).

2.2.6 Meaning for this research: For this research, this means that responsible use of generative AI has to have two characteristics. First, the use has to be visible and explainable, so that students can account for what the tool was used for and which choices they made themselves. Second, the use of AI has to support the learning process rather than replace it. From that idea, the study coach is not a neutral technical tool, but a didactic intervention that aims to strengthen AI literacy, self-regulation and transparent accountability within the context of first-year ICT education at the HvA.

2.3 Didactic guidance and learning mechanisms

For the didactic design of this research, the primer by Dekker is guiding. It distinguishes between free and guided use of AI. That distinction fits this intervention, because the HvA study coach is not designed as an open chat, but as a tool with a bounded coaching role.

Bauer et al. moreover advise looking not only at the tool, but above all at the learning mechanism that is influenced by it. In this research, it is therefore not only about what the chatbot returns, but about the question of whether students keep taking intermediate steps themselves, keep explaining their choices and use AI less as an answer machine.

Within the classification by Bauer et al., the emphasis in this research lies on the student's own effort: what students ultimately show independently. The study coach is therefore not assessed on the quality of the generated output, but on the question of whether its use is associated with more visible and more independent conduct by students.

From that same didactic line, KIES is useful as a concrete educational translation of task-oriented AI guidance. The framework asks teachers and students to weigh up, for each subtask, whether AI supports learning or rather replaces it, and links that consideration to choices about role, instruction, evaluation and ground rules. For this research, that is relevant above all for theory, planning and reflection, where a coaching role can more often be learning-oriented, and for code-oriented tasks, where the division of work has to remain much more tightly bounded (Mutsaarts, 2026).

3. Methodology

3.1 Research design

This research is set up as a practice-based mixed-methods study with a quasi-experimental core. The HvA generative AI study coach is treated in it as a didactic prototype for a teaching method within the teaching practice of the HvA. The core of the design consists of a comparison between two conditions within existing first-year ICT classes: a condition in which students work with a study coach and a condition in which students use generative AI more openly. Because existing classes and regular lesson situations were used, this is not a fully randomised experiment.

The design also has a step-by-step character. The prototype was used, evaluated and, where necessary, sharpened in practice during the research.

To answer the main question systematically, a suitable method was chosen for each sub-question. As a result, the focus is not only on measurable outcomes, but also on the definition of concepts, current use, design conditions, historical trends and the way in which the use of AI can be recorded responsibly.

3.1.1 Overview of sub-questions and methods: In the exploratory phase, a stakeholder analysis was used to order interests, influence and risk for carrying out the research. Students, teachers and supervisors were weighed heavily, because they bear directly on the learning objectives, assessability and feasibility of the intervention. ICTS and tool developers were weighed fairly heavily, because technical feasibility and context links are decisive for use in the HvA environment. Researchers and ethics experts were weighed less heavily, as a source for theoretical interpretation and sharpening.

Alongside this, a risk analysis was drawn up. Risks weighed heavily were the dropout of participants, contamination of the open condition through exchange between groups, and the privacy of the data collected. Risks weighed less heavily concerned delay in the series of measurements and limited adoption of the coach prompt. This weighting was used to sharpen instructions, intervention material and data collection before the comparison in class started.

Besides these exploratory steps, the research is ordered with the DOT framework. That framework helped to treat literature, teaching practice and the prototype of the study coach not in isolation, but always in relation to one another. Within that DOT framework, this research deliberately combines three method types from ICT Research Methods: a Library method in the form of a literature study, Field methods in the form of interviews and observations, and a Lab method in the form of a didactic A/B design, in which two variants of the same approach were placed side by side to compare differences in behaviour, independence and learning process (ICT Research Methods, n.d.).

3.1.2 Literature review and concept analysis: The literature review forms the theoretical basis of the research design and fits within the Library category of ICT Research Methods. This method is relevant above all for sub-question 1 and sub-question 3. For sub-question 1, literature is used to define critical thinking in this thesis as a central concept. It is also worked out how this concept relates to cognitive offloading and mental friction, so that it remains clear, throughout the rest of the research, what exactly is being measured and interpreted.

For sub-question 3, literature is used to translate existing insights about didactic guidance, AI literacy, self-regulation, scaffolding and responsible use of AI into design conditions for teaching practice.

Besides the NPULS framework and the distinction between open and coaching use of AI, the AI Fluency Framework and KIES were included in this research as a lens through which to look at the division of tasks, the assessment of AI output and accountability. These frameworks were not used as a measure of effect, but as a theoretical and didactic lens to interpret design choices and observations in teaching practice (Dakan & Feller, 2025; Mutsaarts, 2026).

3.1.3 Practice exploration through surveys, interviews and observations: For sub-question 2 and sub-question 3, a combination of surveys, interviews, observations and analyses of feedback sessions was chosen. Within ICT Research Methods, interviews and observations fall under the Field category, because these methods are meant to make behaviour, experiences and usage patterns visible in the natural teaching context (ICT Research Methods, z.d.; ICT Research Methods, z.d.).

The observations were carried out as participant observations: during the lessons the researcher was present in the

Sub-question	Method	Data source	Design	Analysis	Outcome
1	Literature review and concept analysis	Academic literature on critical thinking, cognitive offloading and mental friction	Definition of concepts	Thematic comparison of definitions	Working definition of critical thinking
2	Survey, observations and analysis of feedback sessions	First survey, lesson observations and feedback sessions	Practice exploration in BIM, CS and Fusion	Descriptive and thematic analysis	Usage patterns and the experienced role of generative AI
3	Literature review, conversations and practice observations	Academic literature, conversation reports and feedback sessions	Comparison of didactic conditions	Thematic interpretation	Didactic design conditions for the study coach
4	Historical data analysis	Historical grade files 2019-2025	Descriptive trend analysis	Comparison between periods	Cautious interpretation of time relations
5	Quasi-experimental comparison	Baseline measurements, feedback sessions and accountability material	Comparison between open and constrained condition	Descriptive comparison per context	First view of learning performance, independence and visibility of the learning process

classroom and followed students' working process directly. This means that there was no remote recording; instead, the researcher sat beside students and could observe the use of the study coach, the conversation patterns and the independence of students at close range. Each observation session was written up afterwards in fixed notes, ordered by teaching context, date and recurring signals. This way of working connects to the definition of participant observation within the Field category of ICT Research Methods, in which the researcher operates as an involved observer in the research environment (ICT Research Methods, z.d.).

3.1.4 Historical data analysis: For sub-question 4, a descriptive analysis is carried out on available test and grade data from earlier years. Within the repository, historical grade files are available for this from first-year cohorts from 2019 to 2025. With this analysis, it can be examined whether patterns or shifts in learning performance are visible over the period before and during the broader rise of generative AI.

This sub-question has an explicitly exploratory character. The analysis can describe trends and signal co-occurrence, but it does not make it possible to attribute changes in performance straightforwardly to a cause, such as the use of generative AI.

3.1.5 Quasi-experimental comparison (A/B): The quasi-experimental comparison forms the core of the practice-based research and is relevant above all for sub-question 5. Methodologically, this design connects to A/B testing within the Lab category of ICT Research Methods, although this is not about large-scale product optimisation but about a didactic A/B comparison within existing classes (ICT Research Methods, z.d.).

By comparing these conditions within regular teaching practice, it can be examined whether differences become visible in learning performance and independence. The comparison is supported by baseline and follow-up measurements in the form of knowledge exercises on, among other things, UML, BPMN and basic statistics, supplemented with self-reports on perceived use and support. As a result, not only snapshots but also changes in performance and behaviour become visible.

3.2 Research population

The baseline measurement was taken by first-year ICT students. They came from three teaching contexts. Business IT & Management focuses in this research mainly on analysis, BPMN and business-oriented assignments. Cyber Security concerns the programming and basic-statistics context. Fusion concerns a project-based context in which analysis, design, building, testing and accountability come together. These specialisations are not treated as one group, because they are assessed on different content.

For the comparison, a constrained and an open condition were used. In the analysis, the focus was not only on the difference between these two conditions, but also on each specialisation. That was necessary because BIM, Cyber Security and Fusion were not assessed on exactly the same content, so the teaching context plays a part in the interpretation of the results.

Figure 3.1 is not included in this publication version to protect the privacy of participants.

3.3 Data collection and instruments

The data collection combines quantitative and qualitative sources. The quantitative data consists of baseline measurements in the digital learning environment on topics that return within the participating specialisations, such as BPMN, basic statistics and UML diagrams. These tests were taken before students had received theory teaching or AI guidance on these topics. The qualitative data consists of Microsoft Forms surveys, lesson observations during feedback sessions and written-up conversation reports. For the first feedback session a coded set of observations is available; for the second feedback session a combined coding of observations spread across BIM, CS and Fusion was used.

The observations were carried out during the feedback sessions in teaching practice. During these meetings, it was recorded for each group or student what kind of help was asked for, what type of AI output came back, whether intermediate steps remained visible, how references and plagiarism were discussed, and in what way the use of AI was accounted for

in GitLab, issues, vlogs and peer review. The practice notes were written up directly after the lessons and then coded in the coding files of the feedback sessions.

This approach was chosen because cognitive offloading cannot be derived directly from a single source. Test grades show something of knowledge and performance, but not of the way students arrive at their answers. For that reason, this research also looked at descriptions of use, reflections on AI and the visibility of intermediate steps in the working process. The observations were not treated as loose impressions, but as systematic practice notes that were compared, for each teaching context, on recurring and context-specific signals.

3.4 Analysis procedure

The analysis procedure was set up in advance on the basis of the main question, the five sub-questions and the chosen mixed-methods design. In this phase, the focus is not on the conclusion, but on the way the collected data is processed systematically. For that reason, it was first established which variables are distinguished in the analysis and how they are read within this research.

The central independent variable is the teaching condition in which students work with generative AI. In the constrained condition, students were instructed to use generative AI as a study coach that activates their own thinking through steering questions and hints. In the open condition, students could use AI more freely, without the same didactic limits. This choice connects to the A/B design from the research and to the primer by Dekker, which stresses that research into AI in education only remains properly interpretable when the difference between free and guided use of AI is clearly taken into account.

During the intervention, the work involved not only different instructions, but also presentation material that differed per group. Research group A received a presentation in which HvA AI Chat was deliberately set up as a study coach, including persona settings, a coach prompt and instructions for daily check-ins and adding project context. Research group B received a more open introduction to HvA AI Chat, without the same coaching frameworks. These materials are included in the appendices as intervention material (Research group A: study-coach instruction, 23 April 2026; Research group B: open HvA AI Chat instruction, 23 April 2026).

As dependent variables, three analysis categories were distinguished in this research: learning performance, independence and visibility of the learning process. Learning performance is mapped through knowledge exercises, baseline measurements, follow-up measurements and available assessment moments within the teaching contexts involved, including BPMN, UML class diagrams and basic statistics. Independence is linked not only to the final score, but also to the extent to which a student formulates choices, reasoning and intermediate steps themselves, without fully worked-out AI output. The visibility of the learning process concerns the extent to which the use of AI is recorded traceably and accountably in surveys, reflections, feedback material and GitLab-related documentation. That choice connects to the primer by Dekker and to the conversation with Ron Mathot, in which transparency, references and clear accountability of AI use were linked directly to assessability and the prevention of plagiarism claims (Mathot, 2026) (see Appendix H1).

Programme, class, type of assignment and measurement moment are included in the analysis as context variables. That is necessary because BIM, CS and Fusion were not assessed on exactly the same content and do not work in identical lesson situations either. For that reason, quantitative outcomes are not combined into one general total score, but are always read and compared per teaching context.

The quantitative analysis starts with ordering baseline measurements, follow-up measurements and available test data per class, programme and condition. Then, for each measurement moment, descriptive statistics are used, in particular the mean and the spread, to make the starting situation, development and mutual differences visible. This step is meant to describe patterns within the same teaching context. The analysis is therefore not set up as proof for a general claim across all programmes. The qualitative analysis uses surveys, interviews, observations, feedback sessions and diary notes as additional process material. This material is coded thematically on recurring patterns in the purpose of use, perceived guidance, dependence, checking of AI output, perceived added value, resistance to guidance and the need for context links with school systems. These themes were not chosen at random, but follow from earlier feedback sessions, from the research diary and from the problem definition around critical thinking, AI literacy and responsible use.

Concepts such as cognitive offloading and mental friction are not treated in this phase as directly measured scales. They function as an analytical lens in the interpretation of observable signals in the data. Cognitive offloading is taken to occur in the analysis when students outsource thinking steps, take over answers or use AI outcomes without their own explanation or check. Mental friction is recognised, by contrast, when students still have to substantiate choices themselves, formulate intermediate steps, correct mistakes or explain their working method. With this way of measuring, the analysis stays close to the actual data and it is avoided that theoretical concepts are presented more precisely than the measurement design allows.

In the final analysis step, the quantitative and qualitative outcomes were placed side by side. The focus was on how performance data, process data and accountability material complement one another within the same research question. This step is not meant to establish effectiveness here, but to make visible how the different data sources are read together.

3.5 Validity, reliability and the limits of the measurement instruments

The measurement instruments used differ in purpose and evidential strength. The first survey is useful for making usage patterns, perceived affirmation and attitude towards the study

coach visible, but not for making general effect claims about learning performance.

That limit also matters for the research design. In this research, the study coach is therefore treated as a learning intervention and not as a marking instrument: the interpretation of student work, conversations and accountability material remains manual and not AI-driven. That connects to the HvA line that AI is not a suitable route for summative assessment (Hogeschool van Amsterdam, z.d.).

The baseline measurements describe the starting level per teaching context and make differences between BIM, Cyber Security and Fusion visible. They do not, however, show an intervention effect, because they precede the coaching or open use of generative AI.

The qualitative coding of feedback sessions and conversations makes behavioural patterns, accountability and differences between contexts visible. At the same time, this source remains interpretive and dependent on the quality of observation, reporting and coding.

The historical grade data is suitable for describing trends and time relations, but not for establishing a causal connection with the rise of generative AI. Changes in cohort composition, forms of assessment and curriculum may also play a part in it.

Conversations with people involved provide context and interpretation for assessability, learning objectives and the organisation of teaching. This source is therefore used as an interpretive addition and not as independent proof for effect claims.

In the editorial finishing of this report, LanguageTool (https://languagetool.org/) was used as a spelling and structure aid. LanguageTool does not generate text and does not produce substantive output; the tool only checks spelling and sentence structure. All substantive choices, wordings and conclusions are the researcher's own. In addition, HvA AI Chat (Hogeschool van Amsterdam, 2026) was used as a writing aid to gather ideas and explore the structure of chapters. The running text was not written by this tool; every passage was drawn up and checked by the researcher.

4. Sub-question 1: Definition of concepts

4.1 Introduction

This first sub-question focuses on the definition of concepts. The question was answered through a literature review of critical thinking, cognitive offloading and mental friction. The concepts were then used as an analytical lens for the interpretation of the practice data.

4.2 Analysis

Within this research, critical thinking is not understood as avoiding generative AI, but as dealing with AI output in an active, reflective and checkable way. A student shows critical thinking when intermediate steps remain visible, choices are substantiated and AI answers are not taken over without checking. Risko and Gilbert describe cognitive offloading as the outsourcing of mental processes to external aids. In the context of this research, this concerns the shift of thinking steps to the tool without the student still actively steering or checking. Mental friction is understood in this study as the effort needed to keep one's own reasoning going, in line with Bjork and Bjork, who show that some degree of effort in fact contributes to learning. That approach connects to the practice finding that affirming use of AI in particular increases the risk that thinking quietly shifts to the tool, while hints, counter-questions and writing out steps in fact produce more mental friction and visibility (see Appendix G) (Zimmerman, 2002).(Bauer, Greiff, Graesser, Scheiter, & Sailer, 2025)

4.3 Sub-conclusion

In this research, critical thinking means that the student keeps the thinking visibly in their own hands. Cognitive offloading arises when that thinking shifts to the tool without clear checking or accountability.

5. Sub-question 2: Use of generative AI in learning practice

5.1 Introduction

This sub-question was answered with the first survey, lesson observations, feedback sessions and additional conversation reports from BIM, Cyber Security and Fusion.

5.2 Description of the research group

The research was carried out within first-year ICT groups in three teaching contexts: Business IT & Management, Cyber Security and Fusion. The data collection consisted of several measurement moments, including a first survey about the use of the study coach, baseline measurements on subject content, feedback sessions in class, written-up conversations and additional accountability material from practice.

The dataset was processed for the three specialisations. For BIM and Fusion, the material comes from the surveys, feedback sessions, conversations and the baseline and final measurements. For Cyber Security, the additional material comes from the second feedback session and the June measurement. The research diary shows that the responses mentioned in this chapter relate only to the first survey about the study coach and therefore not to the whole research.

5.3 Results: first survey on study-coach use

On the basis of the first survey, a first quantitative picture emerges of the use of generative AI and the study-coach approach in teaching practice. The research diary shows that some of these students reported using AI weekly or daily. A small group reported using the study coach proactively, while a larger group reported having tried the approach (see Appendix F1 and Appendix B). Figure 5.1 shows the numbers at group level.

In the same survey, a large group of students reported that the AI often responded too affirmingly in practice. A clear majority described that the chatbot mostly agreed with them or only became critical when they asked for it specifically. In addition, a large group reported that privacy or a safe HvA

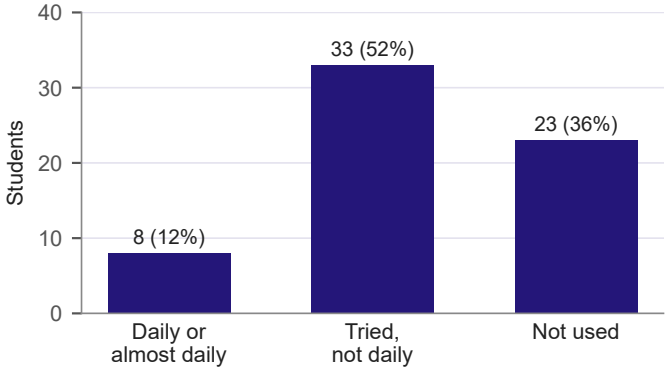


Figure 5.1. Use of the study coach method in the first survey (n = 64).

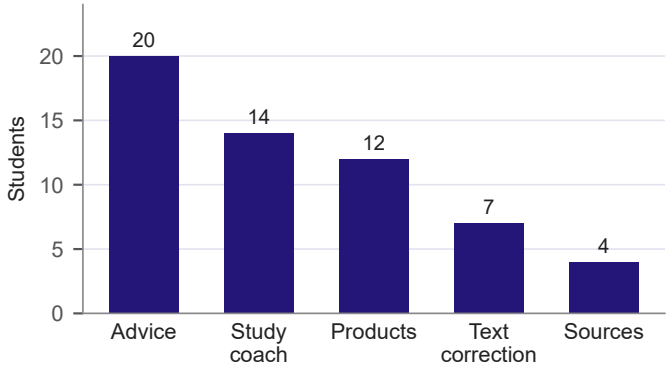


Figure 5.2. Use of generative AI per category in feedback session 1 (n = 36); students could name more than one category.

environment made no visible difference to their use of AI (see Appendix F1 and Appendix B).

These outcomes relate only to the first survey moment about the study coach. They should therefore not be read as a summary of all measurement moments within the research.

5.4 Observations from feedback sessions and conversations

In the lessons and in the first feedback, it emerged that the tool did support students, but also often responded affirmingly. Students described the chatbot as accommodating and not very critical. In the research diary, this pattern was later linked to the observation that a large share of students reported that AI often agreed with them or only became critical when they asked for it specifically (see Appendix B).

The first written-up feedback sessions show that students in BIM classes B101 and B103 used the study coach mainly for check-ins, reflection, improving text and explaining concepts (see Appendix G). Within the same feedback sessions, it was also mentioned that the chatbot sometimes gave unreliable sources or went along too easily with students' reasoning (see Appendix G).

In the CS context, the feedback noted that the same set-up, for coding questions, often led quickly to complete solutions instead of hints or steering questions (see Appendix G). In later reviews of CS and Fusion, a practical usage pattern also returned, in which students used generative AI mainly for explanation, debugging, feedback, improving text and working faster. It also became visible that HvA AI Chat or the study coach was used less readily when students already had their own working method with commercial tools (see Appendix G).

For Fusion and parts of Cyber Security, additional material is available from conversations and feedback sessions (Bleeker, 2026; Bonsema, 2026; Flinkenflögel, 2026; Lafeber, 2026; Mathot, 2026). This material is worked out in Appendix G, Qualitative research data moment 2, and in Appendix H, Interview and conversation reports. The internal source Conversations feedback session 2, all results, is therefore always linked in the text to the relevant appendix or sub-appendix.

5.5 Analysis

The results show that generative AI is already firmly woven into daily learning practice. In the first survey, some of the students surveyed reported using AI weekly or daily. That use is above all practical: for explanation, improving text, planning, feedback and working faster. The second feedback session confirms that pattern in several contexts. In BIM, AI is mentioned mainly for language support, reflection, clarification of assignments and planning. In Cyber Security, use happens more often around debugging, explaining code and finding a direct solution. In Fusion, AI returns in project tasks such as structuring, design choices and reflection.

5.6 Sub-conclusion

First-year ICT students already use generative AI broadly in their daily learning, but mostly in a practical, task-oriented way. The study coach is used, but does not automatically become the dominant working method.

6. Sub-question 3: Didactic conditions for a constrained AI coach

6.1 Introduction

This sub-question was answered by combining literature on didactic guidance, AI literacy and self-regulation with conversations, feedback sessions and practice observations.

6.2 How the study coach was developed and used

The study coach did not come about in its final form all at once. In the first practice tests, the work used ChatGPT and a broad instruction for daily check-ins. The research diary shows that this phase already experimented with coaching use, but that the interaction remained open and depended strongly on how students prompted themselves (see Appendix B).

On 3 March 2026, the set-up was sharpened. From that moment, HvA AI Chat was used with an A/B design, in which group A received a constrained coaching instruction based on counter-questions and group B kept an open instruction (see Appendix B). That change did not alter the underlying tool, but it did alter the didactic role the tool was given in class.

6.3 Interpretation of the findings

The strongest indications around the study coach lie in the learning process. In the qualitative data, the coaching condition returned more often with reflection, explaining concepts, check-ins and working out intermediate steps. That pattern was clearest in assignments that already called for explanation, planning or accountability, such as in BIM and parts of Fusion. For coding questions in CS, the same set-up worked less stably. There, the interaction shifted more quickly towards generating solutions instead of guiding the thinking process. The final measurement in Chapter 7 also shows that the use of the coach, within the same specialisation, goes together with an equal or higher score. This picture indicates that it is not the presence of AI in itself that is decisive, but the combination of type of task, didactic guidance and the extent to which students have to explain their choices (see Appendix G) (NPULS, 2025; Risko & Gilbert, 2016).

This interpretation is directly relevant to the original problem definition. The central problem for the HvA was not that students use generative AI, but that, without limits, it becomes less visible which thinking steps, choices and justifications still genuinely come from the student. The results support that formulation. Open use turned out, in practice, to be associated more often with practical use, speed and a fit with existing commercial tools, while the constrained condition was associated more often with reflection, checking and visibility of the learning process. That does not mean that open use is undesirable by definition, but it does mean that an open setting leads less readily to accountability. In this interim position, the study coach therefore functions above all as a didactic means to slow down the learning process and make it more checkable, not as proof that higher learning performance has already been achieved (see Appendix G) (NPULS, 2025; Risko & Gilbert, 2016).

A second relevant outcome is that, in the current survey and feedback sessions, a safe HvA environment did not emerge as a decisive reason for different AI use. Students valued above all direct usefulness, speed and a fit with their existing workflow. That also explains why several groups expressed a need for links with GitLab, timetable information and other school systems. For students, a safe or institutional environment therefore becomes meaningful above all when that environment also fits the concrete working process better than the tools they already use. This outcome sharpens the HvA issue further: responsible use of AI cannot be enforced solely through infrastructure or policy, but also has to be embedded didactically in assignments, forms of accountability and assessment moments (see Appendix G) (ICT Research Methods, z.d.; NPULS, 2025).

6.4 Relation to the literature

The outcomes broadly fit the literature that describes generative AI not as automatically positive or negative, but as a means whose effect depends on the form of use and the educational design. Miao and Holmes stress that generative AI in education should not be used neutrally, but should be pedagogically designed. The current results connect to that. It was precisely in the constrained condition, where use was bounded through a coaching role, that it became visible more often that students had to explain what they were doing and how they arrived at a choice. That fits the idea that the learning value of AI lies not only in the system's output, but in the way the interaction elicits mental activity or, conversely, replaces it (Miao & Holmes, 2023; Risko & Gilbert, 2016).

The contrast between cognitive offloading and learning-oriented support is also filled in further by the results. Risko and Gilbert describe cognitive offloading as the outsourcing of mental processes to external aids. In this research, it became visible that this shift occurs not only when students literally take over text or code, but also when the interaction contains so little contradiction or friction that one's own considerations are barely needed any more. The recurring yes-man effect is relevant in that respect. When the tool mainly affirms, the chance increases that students move on more quickly without checking their own reasoning. At the same time, the same results show that AI does not necessarily lead only to outsourcing. In contexts where students still had to reflect, plan or explain, the same tool could in fact function as support for self-regulation and for making choices explicit. The results therefore support a conditional rather than an absolute view of the use of AI in education (see Appendix G) (Risko & Gilbert, 2016; Zimmerman, 2002).

The results also connect to the distinction between AI use that replaces learning processes and AI use that supports learning processes, as described by Bauer et al. Within this research, that distinction turned out to be not only methodologically useful, but also didactically meaningful. The two conditions did not work with genuinely different technology, but with a different allocation of roles to the same technology. It was precisely because of this that it became visible that small differences in instruction and the wording of an assignment could be associated with different usage patterns. At the same time, the research also shows that a coaching prompt is not enough in itself. In CS, the risk remained that the tool still generated complete solutions. This research therefore shows that a coaching use does not work out the same way in every teaching context. The didactic effect turns out to be domain- and task-dependent and calls for further sharpening per subject type and per product form (see Appendix G) (Bauer, Greiff, Graesser, Scheiter, & Sailer, 2025).

The AI Fluency Framework and KIES offer additional language for reading these outcomes. The emphasis on delegation and discernment in Dakan and Feller connects to the question of whether students still deliberately choose, for each part of a task, what they do themselves and how they check output. The KIES translation by Mutsaarts makes that same logic concrete at the subtask level. The current results in particular show that this fine-grained consideration is needed: for explanation, planning and reflection, a coaching use can be learning-oriented, while for code, modelling or other technical work there is more quickly a case of unwanted outsourcing.

The current findings therefore also connect to the HBO-

ICT position paper AI in the curriculum, which distinguishes between AI as a learning and feedback instrument and AI in qualifying situations. That distinction in particular is visible in this research: the constrained study coach is useful above all where explanation, planning and reflection are central, while for tasks that have to show individual mastery or lasting knowledge, extra limits remain necessary (Berends, Flinkenflögel, & Van Swieten, 2026).

6.5 Analysis: the study coach in practice

The current results indicate that a study coach becomes useful above all when several didactic conditions are met at the same time. A first condition is targeted guidance. In BIM classes and in parts of Fusion, the coaching condition was recognised above all when teachers made clear what the role of the tool was and what output was or was not wanted. A second condition is limiting the kind of answer the tool gives. In Cyber Security in particular, the current set-up turns out to be not yet sharp enough, because for coding questions the tool gives complete solutions too quickly instead of hints or counter-questions. A third condition is visible accountability, so that prompts, intermediate steps and checking of output become part of the regular working process.

6.6 Limits of the research

The research aim has thereby been partly achieved. The research makes clearer under what didactic conditions a generative AI study coach can contribute to reflection, accountability and the checking of AI output. The June final measurement shows, per specialisation, a first positive connection between the use of the coach and equal or higher scores. Because of the small groups and the difference in test difficulty for BIM, this remains a cautious connection and not yet firm proof that the intervention leads to durably higher learning performance. The strongest and most stable signals in the dataset remain usage patterns, perceived guidance and the visibility of the learning process (see Appendix G) (Hogeschool van Amsterdam, 2026).

An important strength of the research is that it was carried out within existing first-year classes and regular lesson situations. As a result, the outcomes are directly connected to real teaching practices in BIM, CS and Fusion, instead of to an artificial test situation. Moreover, several data sources were combined: surveys, feedback sessions, conversations and historical grade data. That combination makes it possible to look not only at what students say, but also at what becomes visible in their working method and accountability. For a topic such as cognitive offloading, which is not easy to capture in a score, that triangulation is methodologically relevant (see Appendix G) (Hogeschool van Amsterdam, 2026; ICT Research Methods, z.d.).

Against these strengths stand clear limitations. First, the teaching contexts were not equal in content: BIM, CS and Fusion worked with different assignments, forms of assessment and knowledge domains. That limits the comparability between specialisations and makes it risky to read outcomes as an effect of the intervention. Second, the intervention was sharpened during the research. From the point of view of practice-based research, that is defensible, but it reduces the tightness of the comparison between early and later measurement moments. Third, parts of the data rest on self-report, reflection and qualitative coding. Those sources are valuable in content, but sensitive to social desirability, selective memory and differences in wording. Fourth, the historical grade analysis does show a difference between periods, but no causal connection. The numbers, cohorts and teaching contexts differ too strongly for that. Finally, the fieldwork in particular showed that adoption is determined not only by didactics, but also by the technical maturity of the tool. Part of what became visible in this research therefore says something not only about learning with AI, but also about the current limitations of an HvA application that is not yet fully context-linked (see Appendix G) (Hogeschool van Amsterdam, 2026; ICT Research Methods, z.d.).

The current results shift the question from whether students use AI to under what conditions that use still serves learning, assessability and independence. For this programme, the heart of the matter therefore lies less in banning or allowing, and more in assignments, instructions and forms of accountability that keep visible what students do themselves. The study coach can help in this, but only when it fits the assignment and the student's working process. Where that fit is lacking, use shifts more quickly towards pragmatism, speed and the outsourcing of thinking. This research therefore offers above all a first definition of the conditions under which a study coach can be didactically meaningful (see Appendix G) (NPULS, 2025; Risko & Gilbert, 2016).

6.7 Sub-conclusion

An AI coach succeeds as study support when the tool guides in a targeted way, is designed task-by-task, does not give away complete solutions, fits the teaching and working context and makes use visible.

7. Sub-questions 4 and 5: Performance, independence and visibility of the learning process

7.1 Introduction

This chapter answers two sub-questions that lie close together. Sub-question 4 is answered with a descriptive analysis of historical grade data over the period 2019 to 2025. Sub-question 5 is answered with the quasi-experimental comparison between an open and a constrained condition, supplemented with baseline measurements, feedback sessions and accountability material from practice. The outcomes are worked out further under sections 7.5 to 7.8.

7.2 Analysis: historical grade data

For the historical grade analysis, use was made of one multi-year overview file and seven separate annual overviews over the period 2019 to 2025. These source files are included in the appendices (Hogeschool van Amsterdam, 2026).

The multi-year overview contains a large number of data rows. Each row includes, among other things, the year, the part of the programme, the code, the study guide, the numbers of grades per grade category, the total number of grades and the average grade (Hogeschool van Amsterdam, 2026). Figure 7.1 and Figure 7.2 are not included in this publication version to protect the privacy of participants.

Figure 7.1 is not included in this publication version to protect the privacy of participants.

The separate annual overviews over 2019 to 2025 each contain a cross-table with partial-test data and a sheet with metadata. This means that, for this period, both a multi-year overview and a series of separate annual overviews are available (Hogeschool van Amsterdam, 2026).

On the basis of facts, only a time relation with the rise of publicly available generative AI can be linked to these historical grades. ChatGPT became publicly available on 30 November 2022. In the first survey and the feedback sessions of this research, it became visible that generative AI was already being used in several teaching contexts. Within the first survey, some of the students surveyed reported using AI weekly or daily (see Appendix B). This means that the years 2019 to 2022 fall in the period before the broad public availability of ChatGPT, while 2023 to 2025 fall in the period after it (see Appendix B) (OpenAI, 2022).(OpenAI, 2022)

Figure 7.2 is not included in this publication version to protect the privacy of participants.

This breakdown describes a difference between the period before and after the broad public availability of ChatGPT, but on the basis of this data it establishes only a time relation and not a causal connection (Hogeschool van Amsterdam, 2026; OpenAI, 2022).

7.3 Coherence

In the historical grade data, a difference is visible between the period before and after the broad public availability of ChatGPT. For 2019 to 2022, the multi-year overview contains, on average, higher grades than for several components from 2024 and 2025. That shift cannot, however, be attributed directly to the rise of generative AI. The annual figures do not show a simple linear decline, and cohort composition, forms of assessment and curriculum differ per year. The analysis therefore shows, at most, a cautious time relation.

7.4 Sub-conclusion

The comparison of historical grades shows a difference between the period before and after the broad public availability of ChatGPT. On the basis of this data, no causal connection with generative AI can be established. There are three reasons for this. First, several changes in the teaching coincide in the same period, such as a new first-year design, different forms of assessment and changing teaching teams. Second, the dataset contains no information about whether and how students used AI for the assessed assignments. Third, these are aggregates of cohorts, not individual learning gains per student. The conclusion is therefore that the series of grades shows a pattern that coincides with the rise of generative AI, but that this pattern can also be explained by other educational factors.

7.5 Introduction to sub-question 5

This sub-question was answered through the quasi-experimental comparison between an open and a constrained condition, supplemented with baseline measurements, feedback sessions and accountability material from practice.

7.6 Baseline measurement per teaching context

The baseline measurement shows, per specialisation, differences in starting level, expressed as the percentage of correctly answered questions. These outcomes show that the participating teaching contexts do not start from the same level. Figure 7.3 is not included in this publication version to protect the privacy of participants.

The average starting levels varied between the specialisations. Within Cyber Security, the average on the basic-statistics test was the highest, followed by BIM on the BPMN test and the Fusion groups on the UML questions. For several individual UML questions, the percentage of correct answers was clearly lower (see Appendix F2, F3 and F4). The exact percentages have been left out of this publication version to protect the privacy of participants.

These figures are used in this chapter only as a description of the starting level per teaching context. They function as a reference point for later comparison within the same specialisation and the same condition.

Figure 7.3 is not included in this publication version to protect the privacy of participants.

7.7 Analysis of sub-question 5

Since 1 June 2026, the final measurement has also been taken. As a result, it is now possible to compare, per specialisation, with the same group and the same kind of test. For BIM, students took both the baseline measurement in February and the final measurement in June on the BPMN test. The average rose, although the June test had more questions than the baseline, so this is not a pure effect. For Fusion, the students took both UML tests with exactly the same questions. The group with the study-coach prompt, group A, went up slightly, while the group without that prompt, group B, dropped. For Cyber Security, there is only a measurement in June on the basic-statistics test. Students who reported having used the study coach scored higher on average than students who did not. Figures 7.4 to 7.6 show the group averages; the breakdown per student is not shared, to protect the privacy of participants.

Alongside these findings, the picture from the second feedback session still holds. In Fusion, the study coach was recognised clearly more often in group A than in group B. In BIM, the coaching working method returned more often with reflection, check-ins and naming intermediate steps. The main added value therefore lies in the visibility and checkability of the learning process.

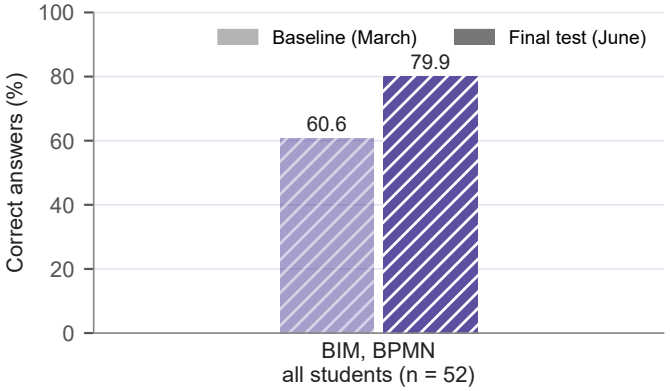


Figure 7.4. BIM, BPMN test: average score at the baseline and the final test, all students (n = 52). The final test had 14 questions instead of 10, so this is not a pure comparison.

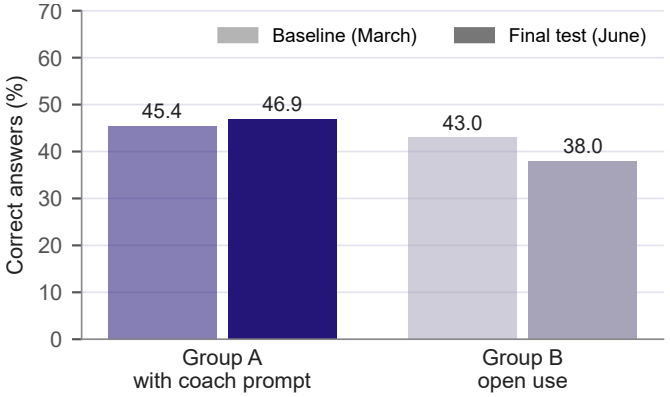


Figure 7.5. Fusion, UML: paired groups A and B from March to June, same 23 students and the same 10 questions.

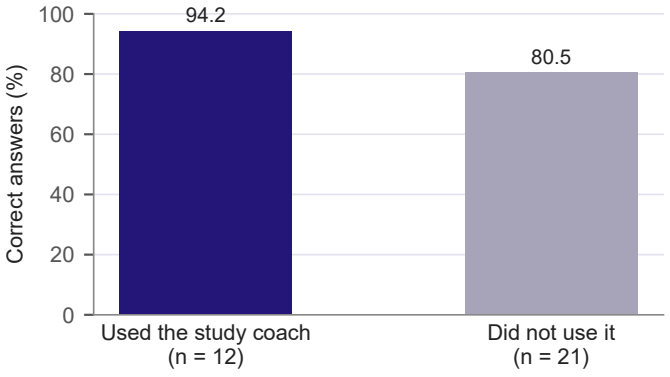


Figure 7.6. Cyber Security, basic statistics in June: average score by self-reported use of the study coach. There was no baseline measurement in this programme.

7.8 Sub-conclusion for sub-question 5

The final measurement of 1 June makes a first comparison per specialisation possible. The picture is mixed, but positive in parts. For BIM, the final score is clearly higher than the baseline. For Fusion, the group with the study-coach prompt holds its level, while the group without that prompt drops. For Cyber Security, students who used the coach score higher than students who did not. Because of the small numbers and the difference in test difficulty for BIM, this remains a cautious statement. The strongest and most stable result is the better visibility of the learning process: more traceable thinking steps, clearer accountability and better checking of AI use.

8. Final conclusion

In the final conclusion below, the five sub-conclusions are first briefly repeated. Then comes the answer to the main question, which rests on these sub-conclusions.

Sub-question 1 (definition of concepts) shows that critical thinking means that the student keeps the thinking visibly in their own hands, and that cognitive offloading arises when that thinking shifts to the tool without checking. Sub-question 2 (use in practice) shows that generative AI is already firmly woven into the learning process of first-year students and that the use is above all practical, with clear differences between BIM, Cyber Security and Fusion. Sub-question 3 (didactic conditions) shows that a study coach works above all when targeted guidance, assignment-focused instruction and visible accountability are present at the same time. Sub-question 4 (historical learning performance) shows that there is a difference between the period before and after the rise of ChatGPT, but that no causal connection can be drawn from it. Sub-question 5 (learning performance) shows that a firm performance effect cannot yet be established at this point, but that independence and the visibility of the learning process do increase.

With those sub-conclusions as a basis, the answer to the main question of this research now follows:

> To what extent does an HvA generative AI study coach help to account transparently for learning performance, without the critical thinking of first-year ICT students beginning to suffer as a result?

On the basis of the current results, the final conclusion is that a constrained HvA generative AI study coach helps above all to make the learning process more transparent. The first final measurement of 1 June also shows that the use of the coach goes together with equal or higher test scores within the same specialisation. Because of the small numbers, that is still a cautious signal and not firm proof. At this point, the added value lies above all in the visibility of thinking steps, clearer accountability and stronger checking of AI output, not in an already proven rise in learning performance.

The study coach therefore supports above all situations in which students have to explain, plan, reflect and make their working method visible. In BIM and parts of Fusion in particular, this fits the nature of the assignments. In Cyber Security, it also turns out that the same approach only works when the tool is bounded more tightly, because otherwise coding questions turn into complete solutions too quickly.

For the main question, this means that transparent accountability and critical thinking do not stand apart from one another, but in fact strengthen one another. As soon as students

have to show their prompts, intermediate steps, choices and checking of AI output more clearly, the chance decreases that thinking shifts invisibly to the tool. The study coach thereby helps above all to bound cognitive offloading better and to strengthen the assessability of the learning process, without a firm performance effect already being demonstrated on the basis of the current series of measurements.

8.1 Reliability and limits

This final conclusion is based on a literature review, a first survey, baseline measurement 1, feedback session 1, feedback session 2, the final measurements of June 2026, conversation reports from people involved and historical grade data. As a result, a broad picture of teaching practice emerges. The conclusion should, however, be read with restraint, because the groups per specialisation are small and the tests differ from one specialisation to another.

8.2 The place of this research in ongoing developments

This research does not stand on its own. Within and beyond the Amsterdam University of Applied Sciences, several projects are under way aimed at the responsible use of generative AI in education. Examples are the HBO-ICT position paper AI in the curriculum, the AI Fluency Framework by Dakan and Feller, the NPULS line and ongoing guides from other universities of applied sciences and universities. Many of those projects focus on similar questions around AI literacy, assessment and didactics (Berends, Flinkenflögel, & Van Swieten, 2026).(2025)

The findings from this research connect to that broader development. The teacher guide delivered in this research adds to that whole by taking the step from general frameworks to a concrete working method for the first-year ICT cohort. At the same time, it is important to read the outcomes of this research alongside the parallel guides appearing within higher professional and university education in 2026. Further research could therefore look at how this guide relates to other developments and where exchange between programmes adds value.

9. Recommendations

The recommendations below build on the proof of concept delivered in this research: the teacher guide Teacher guide - GenAI HvA first year - 2026-2027. This product translates the research findings into concrete guidelines for teachers who work in the combined first year with sprint and project teaching. The recommendations below therefore no longer focus on still having to build a product, but on the introduction, application and further development of this guide within teaching practice (see Appendix I).

9.1 Introduce the teacher guide in a targeted way for theory, planning and reflection

The results show that the coaching use of HvA AI Chat is at this point most useful in teaching contexts where explanation, planning, reflection and understanding are central. In BIM and parts of Fusion in particular, that approach returned more often with check-ins, explaining concepts, structuring and making intermediate steps explicit. For the programme, it therefore makes sense to apply the teacher guide in the 2026-2027 academic year first in a targeted way for these kinds of tasks and teaching blocks, and not yet as a generic approach for all assignments. That choice connects directly to the proof of concept as worked out in the guide (see Appendix I and Appendix G).

9.2 Use the guide as a fixed basis for instruction and teacher supervision

The results show that HvA AI Chat does not automatically lead to responsible or learning-oriented use. The effect is connected to the way the tool is embedded didactically in assignments, instructions and supervision moments. For the HvA, this means that the teacher guide only has value when it is also genuinely used as a shared working method in lesson briefings, sprint starts, check-ins and supervision moments. For each assignment, it should be clear in advance what the role of the tool is, what form of support is and is not allowed, and how students have to account for their use. Without targeted instruction and teacher supervision, use shifts more quickly towards speed and less visible thinking steps (see Appendix I and Appendix G) (Bauer, Greiff, Graesser, Scheiter, & Sailer, 2025).

9.3 Make AI accountability part of the regular working process

The use of AI should not be checked afterwards, but made visible during the working process. The first survey among students, the conversation with Ron Mathot, a conversation partner on fraud, plagiarism and assessment, on 2 March 2026, the conversation with Constance Bleeker, an expert in SIS and Skills at Studio Next, on 3 March 2026, and the analysis of the second feedback session show that assessability and accountability do not follow automatically from the use of a safe HvA environment. It is therefore advisable to make the accountability components from the teacher guide, such as recording prompts, tools used, intermediate steps, reflections and references, a standard part of assignments and forms of assessment. This can be done, for example, through GitLab issues, vlogs, short process notes or fixed moments of reflection in peer review (see Appendix I, H1, H2 and G).

9.4 For technical assignments, work with a tighter variant of the guide

For code-oriented and technically substantive contexts, a generic teacher approach does not yet seem to fit well enough. In Cyber Security and in later diary notes, it returns that students use AI there mainly in a practical way for errors, debugging, quick explanation and testing ideas. At the same time, the feedback sessions indicate that the current coaching set-up often still gives complete solutions too quickly for coding questions (see Appendix G2). On this basis, the advice is to use a tighter variant of the guide for technical assignments, in which the tool is bounded to hints, error

analysis, checking questions and step-by-step feedback instead of complete solutions. In this way, the guide functions not only as a general handover document, but also as a basis that can be sharpened further per teaching situation (see Appendix I, G and B).

9.5 Use the guide as a handover document for the new first year

The internal design of the 2026-2027 first year shows that the programme is working towards a broader first year in which content, organisation and handover have to fit together well. In that context in particular, it is advisable to use the teacher guide as a handover document for teachers who will work with the new sprint and project structure. In this way, the proof of concept gains a place not only in the research, but also in the practical preparation of the teaching. The guide can serve as a basis for team agreements about what HvA AI Chat is, when it is suitable and how students are steered in using it (see Appendix I and Appendix B).

9.6 Anchor AI literacy early in the first year with the same working method

The results show that students already use AI in several teaching contexts, but that good and critical use does not come about by itself. It is therefore advisable to address AI literacy early in the first year. That does not have to be in the form of a separate course, but it should be a recognisable and compulsory part of the curriculum. Attention should be paid to assessing AI output, recognising limitations and hallucinations, making one's own thinking steps visible and accounting responsibly for prompts and tools. The teacher guide can serve as a shared working method for this, so that introduction, lesson delivery and accountability do not become separated from one another (see Appendix I and Appendix B) (NPULS, 2025).

A useful starting point here is to introduce students early to a fixed language for the division of tasks, guidance, evaluation and ground rules, for example through the AI Fluency Framework and its Dutch KIES translation at the student level (Dakan & Feller, 2025; Mutsaarts, 2026).

9.7 Introduce the guide in phases and evaluate per teaching block

A broader introduction of HvA AI Chat within the first year can at this point only be properly justified when that introduction takes place in phases and stays linked to follow-up measurements. The current results offer sufficient grounds to continue with the guide as a practical product, but not yet to legitimise broad introduction with claims about higher learning performance. It therefore makes sense to apply the guide first in specific teaching blocks, sprints or project phases, and then to evaluate it on the way it is used, the visibility of thinking steps and teachers' experiences. Only when follow-up measurements also give more clarity at the level of learning performance is broader scaling-up properly justified (see Appendix I, F2, F3 and F4) (Bauer, Greiff, Graesser, Scheiter, & Sailer, 2025).

9.8 Let the guide connect to the four learning outcomes and to independent study

The conversation with Michiel Flinkenflögel on 8 June 2026 shows that the first year is working towards four learning outcomes that apply to every student: advising, relating, designing and personal leadership. For each learning outcome there are six to eight success criteria, which are also written in terms of behaviour. In a development test, for example, the student has to develop something themselves and explain their choices. The teacher guide connects to this by linking HvA AI Chat to that behaviour: the student shows which steps they take themselves, which choices they make and why. In this way, the coach becomes an aid in demonstrating the success criteria and not a way of skipping them.

A second line from the conversation is that the programme asks more about the why than the how, in line with the position paper of the AI working group approved within NPULS. For the guide, this means that the coach is used above all to support choices, for example when choosing data types or setting up a web application with a database, and not to deliver ready-made code. That fits the tighter variant for technical assignments from recommendation 9.4.

A third line is the shift from instruction to independent study. According to the conversation, a large part of the students do not yet build their own overview of what belongs to a topic, such as with JavaScript. The guide can help with this by using the coach for making summaries, ordering topics and incorporating feedback, and by having students build that overview themselves. In this way, the guide supports the transition to more independent work that the programme has in mind (see Appendix I, H5 and B) (NPULS, 2025; Flinkenflögel, 2026).

10. References

Anghel, E., Braun, H. I., Friedman, A. A., & Baez-Cruz, M. (2021). College students' critical thinking: Assessment and interpretation. *Journal of Higher Education Theory and Practice, 21*(10). doi:10.33423/jhetp.v21i10.4624

Barcaui, A. (2025). ChatGPT as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention. *Social Sciences & Humanities Open, 12,* 102287. doi:10.1016/j.ssaho.2025.102287

Bauer, E., Greiff, S., Graesser, A. C., Scheiter, K., & Sailer, M. (2025). Looking beyond the hype: Understanding the effects of AI on learning. Educational Psychology Review, 37, 45. Retrieved from https://doi.org/10.1007/s10648-025-10020-8

Berends, J., Flinkenflögel, M., & Van Swieten, R. (2026). *AI in het curriculum: richting en uitgangspunten voor onderwijs en toetsing.* Hogeschool van Amsterdam, HBO-ICT.

Bjork, E. L., & Bjork, R. A. (2011). Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning. In M. A. Gernsbacher, R. W. Pew, L. M. Hough, & J. R. Pomerantz (Eds.), Psychology and the real world: Essays illustrating fundamental contributions to society (pp. 56-64). Worth Publishers.

Bleeker, C. (2026, March 3). Conversation about educational innovation, Amsterdam University of Applied Sciences, Amsterdam.

Bonsema, M. (2026, March 25). Conversation about learning objectives and AI in assignments, Amsterdam University of Applied Sciences, Amsterdam.

Dakan, R., & Feller, J. (2025). AI Fluency Framework. Retrieved from Anthropic: https://aifluencyframework.org/

Dekker, I. (2026). A primer for experiments with AI in education [manuscript in preparation].

Flinkenflögel, M. (2026, June 8). Conversation about the new first year and AI in the curriculum, Amsterdam University of Applied Sciences, Amsterdam.

Guo, Y., & Lee, *D. (2023). Leveraging ChatGPT* for enhancing critical thinking skills. *Journal of Chemical Education, 100*(12), 4876-4883. doi:10.1021/acs.jchemed.3c00505
Hogeschool van Amsterdam. (2026). HvA AI Chat [internal AI chat environment].
Hogeschool van Amsterdam. (2026). Internal dataset: Grades HBO-ICT first year 2019-2025.
Hogeschool van Amsterdam. (n.d.). AI en toetsing: Nakijken en feedback geven met AI, HvA regels en richtlijnen (versie 0.9).
Hogeschool van Amsterdam. (n.d.). Discussienota - Verplicht gebruik HvA AI Chat in het onderwijs.
ICT Research Methods. (n.d.). A/B testing. Retrieved from ICT Research Methods: https://ictresearchmethods.nl/lab/a-b-testing/
ICT Research Methods. (n.d.). Interview. Retrieved from ICT Research Methods: https://ictresearchmethods.nl/field/interview
ICT Research Methods. (n.d.). Literature study. Retrieved from ICT Research Methods: https://ictresearchmethods.nl/library/literature-study
ICT Research Methods. (n.d.). Observation. Retrieved from ICT Research Methods: https://ictresearchmethods.nl/field/observation
ICT Research Methods. (n.d.). Root cause analysis. Retrieved from ICT Research Methods:
https://ictresearchmethods.nl/workshop/root-cause-analysis/
ICT Research Methods. (n.d.). The DOT framework. Retrieved from ICT Research Methods: https://ictresearchmethods.nl/dot-framework/
Kahneman, D. (2011). *Thinking, fast and slow.* New York: Farrar, Straus and Giroux.
Kosar, T., Ostojić, D., Liu, Y. D., & Mernik, M. (2024). Computer science education in ChatGPT era: Experiences from an experiment in a programming course for novice programmers. *Mathematics, 12*(3), 463. doi:10.3390/math12030463
Lafeber, P. (2026, March 26). Conversation about the research design and teaching practice, Amsterdam University of Applied Sciences, Amsterdam.
LanguageTool. (2026). LanguageTool [computer software]. Retrieved from LanguageTool: https://languagetool.org/
Lee, H.-Y., Chen, P.-H., Wang, W.-S., Huang, Y.-M., & Wu, T.-T. (2024). Empowering ChatGPT with guidance mechanism in blended learning: Effect of self-regulated learning, higher-order thinking skills, and knowledge construction. International Journal of Educational Technology in Higher Education, 21, article 47. doi:10.1186/s41239-024-00447-4
Li, C., Cui, H., & Hagedorn, L. S. (2026). The cognitive impact of ChatGPT in higher education: A systematic review of critical and creative thinking outcomes. *Computers and Education: Artificial Intelligence, 10,* 100571. doi:10.1016/j.caeai.2026.100571
Mathot, R. (2026, March 2). Conversation about fraud, plagiarism and assessment, Amsterdam University of Applied Sciences, Amsterdam.
Miao, F., & Holmes, W. (2023). Guidance for generative AI in education and research. UNESCO. Retrieved from
https://unesdoc.unesco.org/ark:/48223/pf0000386693
Mutsaarts, M. (2026). Raamwerk KIES: denkraam voor AI vaardigheid van leerlingen en studenten. AI voor Docenten. Retrieved from
https://aivoordocenten.nl/wp-content/uploads/2026/04/raamwerk_KIES_2026-1.pdf
Ng, D. T., Leung, J. K., Chu, K. W., & Qiao, M. S. (2021). AI literacy: Definition, teaching, evaluation and ethical issues. *Proceedings of the Association for Information Science and Technology, 58*(1), 504-509.
NPULS. (2025, May 27). Visie op toetsing, examinering en AI + handreikingen. Retrieved from NPULS:
https://npuls.nl/kennisbank/visie-op-toetsing-examinering-en-ai-handreikingen
OpenAI. (2022, November 30). Introducing ChatGPT. Retrieved from OpenAI: https://openai.com/index/chatgpt/
Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences, 20*(9), 676-688. doi:10.1016/j.tics.2016.07.002
Suriano, R., Plebe, A., Acciai, A., & Fabio, R. A. (2025). Student interaction with ChatGPT can promote complex critical thinking skills. *Learning and Instruction, 95,* 102011.
Whitmore, J. (2009). Coaching for performance: GROWing human potential and purpose. London: Nicholas Brealey Publishing.
Wing, J. M. (2006). Computational thinking. *Communications of the ACM, 49*(3), 33-35.
Wood, D., Bruner, J. S., & Ross, G. (1976). The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry, 17*(2), 89-100.
Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theory Into Practice, 41*(2), 64-70.

Appendix: Glossary

The working definitions below are used consistently throughout this report.

Generative AI (GenAI). A form of artificial intelligence that can generate new content, such as text, code, images or summaries, based on patterns in training data and user input.

Large language model (LLM). A language model trained on large amounts of text that can, on that basis, predict and generate language in response to a prompt.

AI literacy. The ability to understand, apply, evaluate and use AI responsibly, with attention to knowledge, skills, attitude and ethical considerations.

Generative AI study coach. An AI supporter used in this research that does not give complete answers directly, but prompts students towards their own thinking and accountability through counter-questions, hints and step-by-step guidance.

HvA AI Chat. The internal AI environment of the Amsterdam University of Applied Sciences, used in this research as a safe teaching context for the use of AI.

GROW model. A coaching model that structures conversations around goal, current situation, options and a concrete next step or intention to act.

Prompt. The text input or instruction with which a user directs an AI system to generate an answer, analysis or product.

AI output. The content an AI system generates in response to a prompt, such as text, code, feedback, summaries or suggestions.

Hallucination. An answer generated by AI that can seem convincing, but is factually incorrect, incomplete or impossible to verify.

Yes-man effect. A pattern in which a chatbot mainly affirms, contradicts little and asks few counter-questions, so that the user is prompted less towards critical checking.

Check-in. A short review or progress discussion in which a student reflects on planning, progress or sticking points.

Coach adoption. The extent to which students actually use the study coach as intended in the intervention.

Pragmatic use. Use of AI aimed mainly at speed, convenience or direct output, and less at reflection or deeper learning.

Context link. A connection of an AI tool to existing systems or information sources, such as GitLab, course material or timetable information.

Critical thinking. The ability to actively analyse, check, substantiate and, where necessary, correct information, reasoning and AI output, so that the thinking process visibly stays with the student.

Cognitive offloading. The outsourcing of mental processes to external aids, so that less is asked of one's own memory, reasoning or problem-solving ability.

Mental friction. The productive resistance in a learning process that forces a student not to take over an answer

immediately, but to formulate intermediate steps, choices and justifications themselves.

Self-regulation. The process in which students actively steer and monitor their own learning objectives, approach, progress and evaluation.

Metacognition. The ability to follow, assess and, where necessary, adjust one's own thinking, learning and acting.

System 2 thinking. A slow, conscious and analytical way of thinking that is used for complex considerations, reasoning and problem-solving.

Computational thinking. A way of thinking in which problems are analysed and structured so that they can be solved step by step and systematically, often with the help of concepts from computer science.

Scaffolding. Temporary support that helps a student carry out a task they could not yet manage independently without guidance. That support is reduced as the student develops more mastery.

Accountability behaviour. The extent to which a student makes the use of AI visible, explainable and checkable by recording prompts, tools used, intermediate steps, reflection and references.

Transparency of AI use. The extent to which it is traceable, for student, teacher and assessor, how AI was used within a learning or working process.

Assessment integrity. The extent to which assessment remains valid, fair and checkable, even when students use AI during learning or assessment processes.

Mixed-methods research. A research design in which quantitative and qualitative data are deliberately combined to answer the same research question from several angles.

Quasi-experimental comparison. A comparison between groups or conditions without full random assignment, for example because existing classes are used.

Intervention. The deliberate action in teaching practice that is tested in the research; in this thesis, the use of the constrained study coach.

Didactic prototype. A provisional, practice-tested version of an educational solution.

Teaching condition. The specific variant of the teaching design in which students work, for example the constrained or open AI condition.

A/B design. A comparison between two conditions or groups, for example group A with a constrained study coach and group B with more open use of AI.

Independent variable. The factor that the research examines for whether it makes a difference to the outcomes; in this thesis, the teaching condition.

Dependent variable. The outcome measure examined for whether it changes under the influence of the independent variable, such as learning performance or independence.

Context variable. A background factor that can influence the outcomes and is therefore included in the analysis, such as programme, class, type of assignment or measurement moment.

Baseline measurement. The first measurement before or at the start of an intervention, meant to establish the starting level.

Follow-up measurement. A later measurement that makes changes during or after the intervention visible.

Historical data analysis. An analysis of existing data from earlier years to make patterns or trends in performance visible.

Thematic coding. A way of analysing in which qualitative data is ordered into recurring themes or patterns.

Descriptive statistics. A statistical description of data with, for example, counts, means and spread, without directly proving a causal effect.

Triangulation. Combining several data sources or methods to answer a research question more reliably.

Learning performance. The extent to which knowledge or mastery visibly increases between measurement moments, based on baseline, follow-up or final measurements.

Independence. The extent to which a student formulates choices, reasoning and intermediate steps themselves, without a full AI solution.

Constrained use. A form of AI use in which the role of the tool is didactically bounded in advance and the interaction is deliberately steered.

Unconstrained use. A form of AI use in which students decide for themselves how they use the tool, without didactic guidance set in advance.

Independent performance. The performance a student can deliver independently, without the help of AI or other support.

Augmented performance. A performance in which AI offers support while the student keeps control of the thinking process and the end product.

Hybrid performance. A form of performance in which the AI takes over a large part of the work, so that it is less clear what the student masters themselves.

Didactic guidance. The deliberate design of assignments, instructions and AI interaction to elicit certain learning behaviour, such as reflection, self-regulation and accountability.

Learning mechanism. The underlying process through which a didactic intervention has an effect on learning, such as scaffolding, self-regulation, reflection or feedback.

First year (propedeuse). The first phase of a higher professional education programme, usually the first year, in which basic knowledge, study skills and suitability for the programme are central.

Learning outcome. A concrete result in knowledge, skill or attitude that a student has to be able to demonstrate at the end of a learning activity or course unit.

Assessability. The extent to which it remains visible and checkable, for teachers and assessors, what a student has done, understood and substantiated themselves.

Business IT & Management (BIM). The ICT specialisation that focuses on processes, information provision and the link between organisation and IT.

Cyber Security (CS). The ICT specialisation that focuses strongly on programming, software and technical problem-solving.

Fusion. The ICT teaching context or learning route in which different fields come together in an integrated way.

BPMN. Business Process Model and Notation; a standard for representing business processes in diagram form.

UML class diagram. A diagram form from the Unified Modeling Language that describes classes, attributes, methods and relationships in software design.

DLO. Digital learning environment in which course material, assignments and communication for students and teachers come together.

GitLab. A platform for version control, documentation and collaboration, used among other things for code, issues and recording the process.

IDE. Integrated Development Environment; a software environment in which programmers write, test and debug code.

Peer review. A form of feedback in which students assess or discuss each other's work.